

Computer Simulation Problems

SIM Problems identified by the Multisim/PSpice icon are intended to demonstrate the value of using SPICE simulation to verify hand analysis and design, and to investigate important issues such as allowable signal swing and amplifier nonlinear distortion. Instructions to assist in setting up PSpice and Multisim simulations for all the indicated problems can be found in the corresponding files on the website. Note that if a particular parameter value is not specified in the problem statement, you are to make a reasonable assumption.

Section 9.1: The MOS Differential Pair

9.1 For an NMOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 9.2, let $V_{DD} = V_{SS} = 1.0$ V, $k'_n = 0.4$ mA/V², $(W/L)_{1,2} = 10$, $V_{tn} = 0.4$ V, $I = 0.16$ mA, $R_D = 5$ k Ω , and neglect channel-length modulation.

- Find V_{OV} and V_{GS} for each transistor.
- For $V_{CM} = 0$, find V_S , I_{D1} , I_{D2} , V_{D1} , and V_{D2} .
- Repeat (b) for $V_{CM} = +0.4$ V.
- Repeat (b) for $V_{CM} = -0.1$ V.
- What is the highest value of V_{CM} for which Q_1 and Q_2 remain in saturation?
- If current source I requires a minimum voltage of 0.2 V to operate properly, what is the lowest value allowed for V_S and hence for V_{CM} ?

9.2 For the PMOS differential amplifier shown in Fig. P9.2 let $V_{tp} = -0.8$ V and $k'_p W/L = 4$ mA/V². Neglect channel-length modulation.

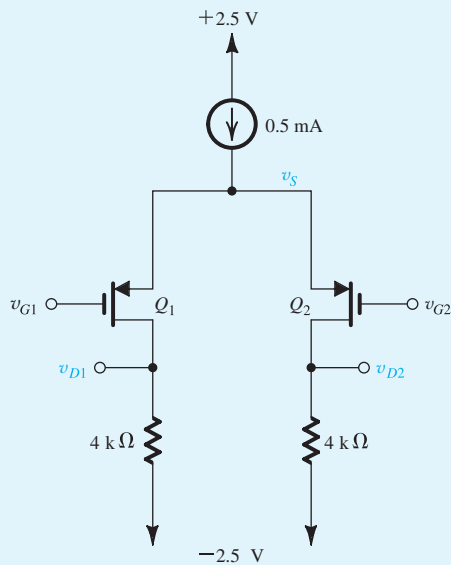


Figure P9.2

- For $v_{G1} = v_{G2} = 0$ V, find $|V_{OV}|$ and V_{SG} for each of Q_1 and Q_2 . Also find V_S , V_{D1} , and V_{D2} .
- If the current source requires a minimum voltage of 0.4 V, find the input common-mode range.

9.3 For the differential amplifier specified in Problem 9.1 let $v_{G2} = 0$ and $v_{G1} = v_{id}$. Find the value of v_{id} that corresponds to each of the following situations:

- $i_{D1} = i_{D2} = 0.08$ mA; (b) $i_{D1} = 0.12$ mA and $i_{D2} = 0.04$ mA;
- $i_{D1} = 0.16$ mA and $i_{D2} = 0$ (Q_2 just cuts off); (d) $i_{D1} = 0.04$ mA and $i_{D2} = 0.12$ mA; (e) $i_{D1} = 0$ mA (Q_1 just cuts off) and $i_{D2} = 0.16$ mA. For each case, find v_S , v_{D1} , v_{D2} , and $(v_{D2} - v_{D1})$.

SIM 9.4 For the differential amplifier specified in Problem 9.2, let $v_{G2} = 0$ and $v_{G1} = v_{id}$. Find the range of v_{id} needed to steer the bias current from one side of the pair to the other. At each end of this range, give the value of the voltage at the common-source terminal and the drain voltages.

9.5 Consider the differential amplifier specified in Problem 9.1 with G_2 grounded and $v_{G1} = v_{id}$. Let v_{id} be adjusted to the value that causes $i_{D1} = 0.09$ mA and $i_{D2} = 0.07$ mA. Find the corresponding values of v_{GS2} , v_S , v_{GS1} , and hence v_{id} . What is the difference output voltage $v_{D2} - v_{D1}$? What is the voltage gain $(v_{D2} - v_{D1})/v_{id}$? What value of v_{id} results in $i_{D1} = 0.07$ mA and $i_{D2} = 0.09$ mA?

D 9.6 Design the circuit in Fig. P9.6 to obtain a dc voltage of +0.1 V at each of the drains of Q_1 and Q_2 when $v_{G1} = v_{G2} = 0$ V. Operate all transistors at $V_{OV} = 0.15$ V

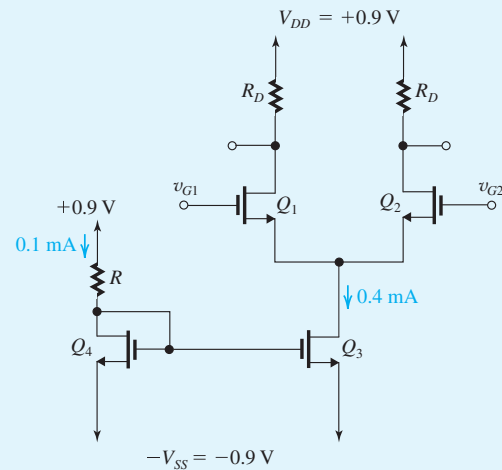


Figure P9.6

and assume that for the process technology in which the circuit is fabricated, $V_m = 0.4$ V and $\mu_n C_{ox} = 400 \mu\text{A}/\text{V}^2$. Neglect channel-length modulation. Determine the values of R , R_D , and the W/L ratios of Q_1 , Q_2 , Q_3 , and Q_4 . What is the input common-mode voltage range for your design?

9.7 The table providing the answers to Exercise 9.3 shows that as the maximum input signal to be applied to the differential pair is increased, linearity is maintained at the same level by operating at a higher V_{OV} . If $|v_{id}|_{\max}$ is to be 220 mV, use the data in the table to determine the required V_{OV} and the corresponding values of W/L and g_m .

9.8 Use Eq. (9.23) to show that if the term involving v_{id}^2 is to be kept to a maximum value of k then the maximum possible fractional change in the transistor current is given by

$$\frac{\Delta I_{\max}}{I/2} = 2\sqrt{k(1-k)}$$

and the corresponding maximum value of v_{id} is given by

$$v_{id\max} = 2\sqrt{k}V_{OV}$$

Evaluate both expressions for $k = 0.01$, 0.1 , and 0.2 .

9.9 A MOS differential amplifier biased with a current source $I = 200 \mu\text{A}$ is found to switch currents completely to one side of the pair when a difference signal $v_{id} = 0.3$ V is applied. At what overdrive voltage will each of Q_1 and Q_2 be operating when $v_{id} = 0$? If v_{id} for full current switching is to be 0.5 V, what must the bias current I be changed to?

D 9.10 Design the MOS differential amplifier of Fig. 9.5 to operate at $V_{OV} = 0.25$ V and to provide a transconductance g_m of 1 mA/V. Specify the W/L ratios and the bias current. The technology available provides $V_t = 0.5$ V and $\mu_n C_{ox} = 400 \mu\text{A}/\text{V}^2$.

9.11 For the MOS differential pair in Fig. 9.5, specify the value of $v_{id} \equiv v_{G1} - v_{G2}$, in terms of V_{OV} , that

- causes i_{D1} to increase by 10% above its equilibrium value of $I/2$.
- makes $i_{D1}/i_{D2} = 1.0; 2.0; 1.1; 1.01; 20$.

9.12 An NMOS differential amplifier is operated at a bias current I of 0.2 mA and has a W/L ratio of 32 , $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $V_A = 10$ V, and $R_D = 10$ k Ω . Find V_{OV} , g_m , r_o , and A_d .

D 9.13 It is required to design an NMOS differential amplifier to operate with a differential input voltage that

can be as high as 0.1 V while keeping the nonlinear term under the square root in Eq. (9.23) to a maximum of 0.04 . A transconductance g_m of 2 mA/V is needed and the amplifier is required to provide a differential output signal of 1 V when the input is at its maximum value. Find the required values of V_{OV} , I , R_D , and W/L . Assume that the technology available has $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$ and $\lambda = 0$.

D 9.14 Design a MOS differential amplifier to operate from ± 1 -V power supplies and dissipate no more than 1 mW in the equilibrium state. The differential voltage gain A_d is to be 10 V/V and the output common-mode dc voltage is to be 0.2 V. (Note: This is the dc voltage at the drains.) Assume $\mu_n C_{ox} = 400 \mu\text{A}/\text{V}^2$ and neglect the Early effect. Specify I , R_D , and W/L .

D 9.15 Design a MOS differential amplifier to operate from ± 1 -V supplies and dissipate no more than 1 mW in its equilibrium state. Select the value of V_{OV} so that the value of v_{id} that steers the current from one side of the pair to the other is 0.25 V. The differential voltage gain A_d is to be 10 V/V. Assume $k'_n = 400 \mu\text{A}/\text{V}^2$ and neglect the Early effect. Specify the required values of I , R_D , and W/L .

9.16 An NMOS differential amplifier employing equal drain resistors, $R_D = 47$ k Ω , has a differential gain A_d of 20 V/V.

- What is the value of g_m for each of the two transistors?
- If each of the two transistors is operating at an overdrive voltage $V_{OV} = 0.2$ V, what must the value of I be?
- For $v_{id} = 0$, what is the dc voltage across each R_D ?
- If v_{id} is 20 -mV peak-to-peak sine wave applied in a balanced manner but superimposed on $V_{CM} = 0.5$ V, what is the peak of the sine-wave signal at each drain?
- What is the lowest value that V_{DD} must have to ensure saturation-mode operation for Q_1 and Q_2 at all times? Assume $V_t = 0.5$ V.

9.17 A MOS differential amplifier is designed to have a differential gain A_d equal to the voltage gain obtained from a common-source amplifier. Both amplifiers utilize the same values of R_D and supply voltages, and all the transistors have the same W/L ratios. What must the bias current I of the differential pair be relative to the bias current I_D of the CS amplifier? What is the ratio of the power dissipation of the two circuits?

9.18 A differential amplifier is designed to have a differential voltage gain equal to the voltage gain of a common-source amplifier. Both amplifiers use the same values of R_D and supply voltages and are designed to dissipate equal amounts of power in their equilibrium or quiescent state. As well, all the transistors use the same channel length. What must the width W of the differential-pair transistors be relative to the width of the CS transistor?

D 9.19 Figure P9.19 shows a MOS differential amplifier with the drain resistors R_D implemented using diode-connected PMOS transistors, Q_3 and Q_4 . Let Q_1 and Q_2 be matched, and Q_3 and Q_4 be matched.

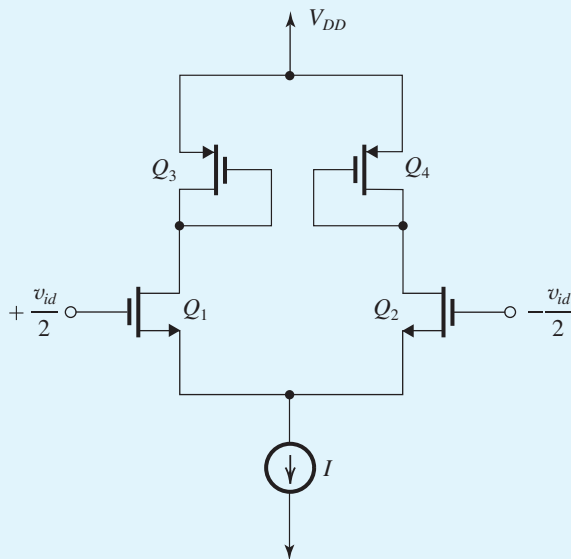


Figure P9.19

- Find the differential half-circuit and use it to derive an expression for A_d in terms of $g_{m1,2}$, $g_{m3,4}$, $r_{o1,2}$, and $r_{o3,4}$.
- Neglecting the effect of the output resistances r_o , find A_d in terms of μ_n , μ_p , $(W/L)_{1,2}$ and $(W/L)_{3,4}$.
- If $\mu_n = 4\mu_p$ and all four transistors have the same channel length, find $(W_{1,2}/W_{3,4})$ that results in $A_d = 10$ V/V.

9.20 Find the differential half-circuit for the differential amplifier shown in Fig. P9.20 and use it to derive an expression for the differential gain $A_d \equiv v_{od}/v_{id}$ in terms of g_m , R_D , and R_s . Neglect the Early effect. What is the gain with

$R_s = 0$? What is the value of R_s (in terms of $1/g_m$) that reduces the gain to half this value?

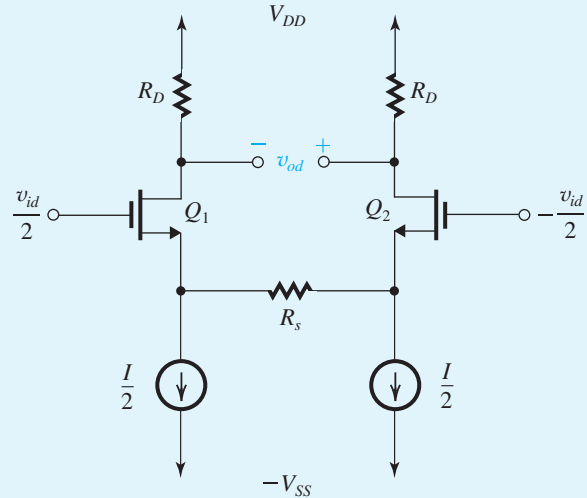


Figure P9.20

***9.21** The resistance R_s in the circuit of Fig. P9.20 can be implemented by using a MOSFET operated in the triode region, as shown in Fig. P9.21. Here Q_3 implements R_s , with the value of R_s determined by the voltage V_c at the gate of Q_3 .

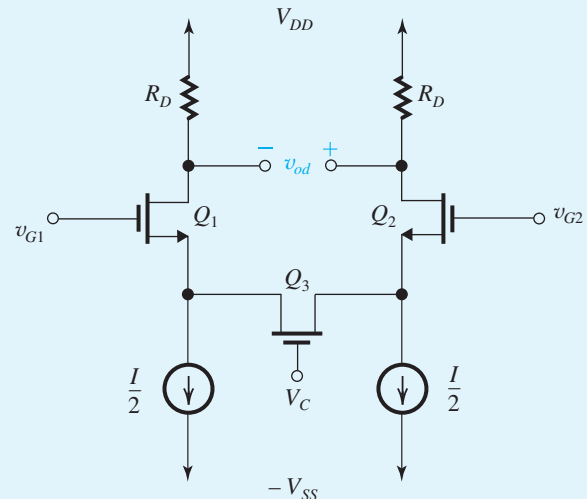


Figure P9.21

- With $v_{G1} = v_{G2} = 0$ V, and assuming that Q_1 and Q_2 are operating in saturation, what dc voltages appear

- at the sources of Q_1 and Q_2 ? Express these in terms of the overdrive voltage V_{OV} at which each of Q_1 and Q_2 operates, and V_t .
- (b) For the situation in (a), what current flows in Q_3 ? What overdrive voltage V_{OV3} is Q_3 operating at, in terms of V_C , V_{OV} , and V_t ?
- (c) Now consider the case $v_{G1} = +v_{id}/2$ and $v_{G2} = -v_{id}/2$, where v_{id} is a small signal. Convince yourself that Q_3 now conducts current and operates in the triode region with a small v_{DS} . What resistance r_{DS} does it have, expressed in terms of the overdrive voltage V_{OV3} at which it is operating? This is the resistance R_s . Now if all three transistors have the same W/L , express R_s in terms of V_{OV} , V_{OV3} , and $g_{m1,2}$.
- (d) Find V_{OV3} and hence V_C that result in (i) $R_s = 1/g_{m1,2}$; (ii) $R_s = 0.5/g_{m1,2}$.

***9.22** The circuit of Fig. P9.22 shows an effective way of implementing the resistance R_s needed for the circuit in Fig. P9.20. Here R_s is realized as the series equivalent of two MOSFETs Q_3 and Q_4 that are operated in the triode region, thus, $R_s = r_{DS3} + r_{DS4}$. Assume that Q_1 and Q_2 are matched and operate in saturation at an overdrive voltage V_{OV} that corresponds to a drain bias current of $I/2$. Also, assume that Q_3 and Q_4 are matched.

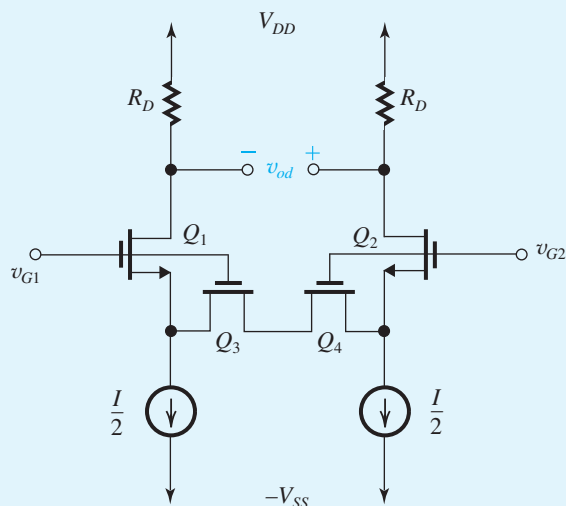


Figure P9.22

- (a) With $v_{G1} = v_{G2} = 0$ V, what dc voltages appear at the sources of Q_1 and Q_2 ? What current flows through Q_3

- and Q_4 ? At what overdrive voltages are Q_3 and Q_4 operating? Find an expression for r_{DS} for each of Q_3 and Q_4 and hence for R_s in terms of $(W/L)_{1,2}$, $(W/L)_{3,4}$, and $g_{m1,2}$.
- (b) Now with $v_{G1} = v_{id}/2$ and $v_{G2} = -v_{id}/2$, where v_{id} is a small signal, find an expression of the voltage gain $A_d \equiv v_{od}/v_{id}$ in terms of $g_{m1,2}$, R_D , $(W/L)_{1,2}$, and $(W/L)_{3,4}$.

D *9.23 Figure P9.23 shows a circuit for a differential amplifier with an active load. Here Q_1 and Q_2 form the differential pair, while the current source transistors Q_4 and Q_5 form the active loads for Q_1 and Q_2 , respectively. The dc bias circuit that establishes an appropriate dc voltage at the drains of Q_1 and Q_2 is not shown. It is required to design the circuit to meet the following specifications:

- (a) Differential gain $A_d = 50$ V/V.
 (b) $I_{REF} = I = 200$ μ A.
 (c) The dc voltage at the gates of Q_6 and Q_3 is $+0.8$ V.
 (d) The dc voltage at the gates of Q_7 , Q_4 , and Q_5 is -0.8 V.

The technology available is specified as follows: $\mu_n C_{ox} = 2.5$ μ A/V²; $V_{tn} = |V_{tp}| = 0.5$ V, $V_{An} = |V_{Ap}| = 10$ V. Specify the required value of R and the W/L ratios for all transistors. Also specify I_D and $|V_{GS}|$ at which each transistor is operating. For dc bias calculations you may neglect channel-length modulation.

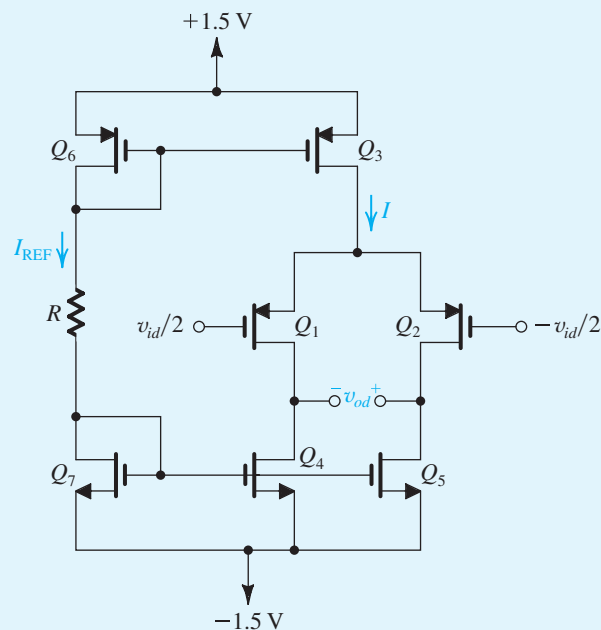


Figure P9.23

***9.24** A design error has resulted in a gross mismatch in the circuit of Fig. P9.24. Specifically, Q_2 has twice the W/L ratio of Q_1 . If v_{id} is a small sine-wave signal, find:

- I_{D1} and I_{D2} .
- V_{OV} for each of Q_1 and Q_2 .
- The differential gain A_d in terms of R_D , I , and V_{OV} .

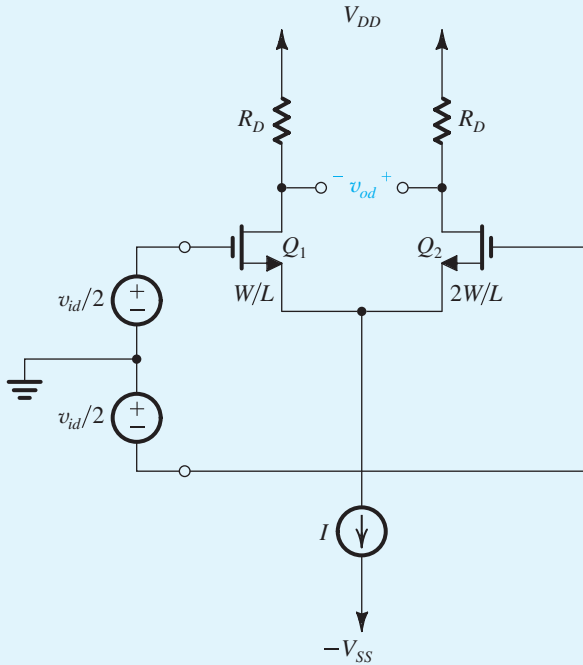


Figure P9.24

D 9.25 For the cascode differential amplifier of Fig. 9.13(a), show that if all transistors have the same channel length and are operated at the same $|V_{OV}|$ and assuming that $V'_{An} = |V'_{Ap}| = |V'_A|$, the differential gain A_d is given by

$$A_d = 2(|V'_A|/|V_{OV}|)^2$$

Now design the amplifier to obtain a differential gain of 500 V/V. Use $|V_{OV}| = 0.2$ V. If $|V'_A| = 5$ V/ μm , specify the required channel length L . If g_m is to be as high as possible but the power dissipation in the amplifier (in equilibrium) is to be limited to 0.5 mW, what bias current I would you use? Let $V_{DD} = V_{SS} = 0.9$ V.

Section 9.2: The BJT Differential Pair

9.26 For the differential amplifier of Fig. 9.15(a) let $I = 0.4$ mA, $V_{CC} = V_{EE} = 2.5$ V, $V_{CM} = -1$ V, $R_C = 5$ k Ω , and

$\beta = 100$. Assume that the BJTs have $v_{BE} = 0.7$ V at $i_C = 1$ mA. Find the voltage at the emitters and at the outputs.

9.27 An *npn* differential amplifier with $I = 0.4$ mA, $V_{CC} = V_{EE} = 2.5$ V, and $R_C = 5$ k Ω utilizes BJTs with $\beta = 100$ and $v_{BE} = 0.7$ V at $i_C = 1$ mA. If $v_{B2} = 0$, find V_E , V_{C1} , and V_{C2} obtained with $v_{B1} = +0.5$ V, and with $v_{B1} = -0.5$ V. Assume that the current source requires a minimum of 0.3 V for proper operation.

9.28 An *npn* differential amplifier with $I = 0.4$ mA, $V_{CC} = V_{EE} = 2.5$ V, and $R_C = 5$ k Ω utilizes BJTs with $\beta = 100$ and $v_{BE} = 0.7$ V at $i_C = 1$ mA. Assuming that the bias current is obtained by a simple current source and that all transistors require a minimum v_{CE} of 0.3 V for operation in the active mode, find the input common-mode range.

9.29 Repeat Exercise 9.7 for an input of -0.3 V.

9.30 An *npn* differential pair employs transistors for which $v_{BE} = 690$ mV at $i_C = 1$ mA, and $\beta = 50$. The transistors leave the active mode at $v_{CE} \leq 0.3$ V. The collector resistors $R_C = 82$ k Ω , and the power supplies are ± 1.2 V. The bias current $I = 20$ μA and is supplied with a simple current source.

- For $v_{B1} = v_{B2} = V_{CM} = 0$ V, find V_E , V_{C1} , and V_{C2} .
- Find the input common-mode range.
- If $v_{B2} = 0$, find the value of v_{B1} that increases the current in Q_1 by 10%.

9.31 Consider the BJT differential amplifier when fed with a common-mode voltage V_{CM} as shown in Fig. 9.15(a). As is often the case, the supply voltage V_{CC} may not be pure dc but might include a ripple component v_r of small amplitude and a frequency of 120 Hz (see Section 4.5). Thus the supply voltage becomes $V_{CC} + v_r$. Find the ripple component of the collector voltages, v_{C1} and v_{C2} , as well as of the difference output voltage $v_{od} \equiv v_{C2} - v_{C1}$. Comment on the differential amplifier response to this undesirable power-supply ripple.

D 9.32 Consider the differential amplifier of Fig. 9.14 and let the BJT β be very large:

- What is the largest input common-mode signal that can be applied while the BJTs remain comfortably in the active region with $v_{CB} = 0$?
- If the available power supply V_{CC} is 2.0 V, what value of IR_C should you choose in order to allow a common-mode input signal of ± 1.0 V?
- For the value of IR_C found in (b), select values for I and R_C . Use the largest possible value for I subject to the

constraint that the base current of each transistor (when I divides equally) should not exceed $2\text{ }\mu\text{A}$. Let $\beta = 100$.

9.33 To provide insight into the possibility of nonlinear distortion resulting from large differential input signals applied to the differential amplifier of Fig. 9.14, evaluate the normalized change in the current i_{E1} , $\Delta i_{E1}/I = (i_{E1} - (I/2))/I$, for differential input signals v_{id} of 2, 5, 8, 10, 20, 30, and 40 mV. Provide a tabulation of the ratio $(\Delta i_{E1}/I)/v_{id}$, which represents the proportional transconductance gain of the differential pair, versus v_{id} . Comment on the linearity of the differential pair as an amplifier.

D 9.34 Design the circuit of Fig. 9.14 to provide a differential output voltage (i.e., one taken between the two collectors) of 1 V when the differential input signal is 10 mV. A current source of 1 mA and a positive supply of +5 V are available. What is the largest possible input common-mode voltage for which operation is as required? Assume $\alpha \simeq 1$.

***9.35** For the circuit in Fig. 9.14, assuming $\alpha = 1$ and $IR_C = 5\text{ V}$, use Eqs. (9.48) and (9.49) to find i_{C1} and i_{C2} , and hence determine $v_{od} = v_{C2} - v_{C1}$ for input differential signals $v_{id} \equiv v_{B1} - v_{B2}$ of 2 mV, 5 mV, 10 mV, 15 mV, 20 mV, 25 mV, 30 mV, 35 mV, and 40 mV. Plot v_{od} versus v_{id} , and hence comment on the amplifier linearity. As another way of visualizing linearity, determine the gain (v_o/v_{id}) versus v_{id} . Comment on the resulting gain graph.

9.36 In a differential amplifier using a 1.5-mA emitter bias current source, the two BJTs are not matched. Rather, one has twice the emitter junction area of the other. For a differential input signal of zero volts, what do the collector currents become? What difference input is needed to equalize the collector currents? Assume $\alpha = 1$.

9.37 This problem explores the linearization of the transfer characteristics of the differential pair achieved by including emitter-degeneration resistances R_e in the emitters (see Fig. 9.17). Consider the case $I = 200\text{ }\mu\text{A}$ with the transistors exhibiting $v_{BE} = 690\text{ mV}$ at $i_C = 1\text{ mA}$ and assume $\alpha \simeq 1$.

- With no emitter resistances R_e , what value of V_{BE} results when $v_{id} = 0$?
- With no emitter resistances R_e , use the large-signal model to find i_{C1} and i_{C2} when $v_{id} = 20\text{ mV}$.
- Now find the value of R_e that will result in the same i_{C1} and i_{C2} as in (b) but with $v_{id} = 200\text{ mV}$. Use the large-signal model.

- Calculate the effective transconductance G_m as the ratio of the difference current, $(i_{C1} - i_{C2})$, to v_{id} in the cases without and with the R_e 's. By what factor is G_m reduced? How does this factor relate to the increase in v_{id} ? Comment.

9.38 A BJT differential amplifier uses a 400- μA bias current. What is the value of g_m of each device? If β is 160, what is the differential input resistance?

D 9.39 Design the basic BJT differential amplifier circuit of Fig. 9.18 to provide a differential input resistance of at least $20\text{ k}\Omega$ and a differential voltage gain of 100 V/V . The transistor β is specified to be at least 100. Specify I and R_C .

9.40 For a differential amplifier to which a total difference signal of 10 mV is applied, what is the equivalent signal to its corresponding CE half-circuit? If the emitter current source I is $200\text{ }\mu\text{A}$, what is r_e of the half-circuit? For a load resistance of $10\text{ k}\Omega$ in each collector, what is the half-circuit gain? What magnitude of signal output voltage would you expect at each collector? Between the two collectors?

9.41 A BJT differential amplifier is biased from a 0.5-mA constant-current source and includes a 400- Ω resistor in each emitter. The collectors are connected to V_{CC} via 10-k Ω resistors. A differential input signal of 0.1 V is applied between the two bases.

- Find the signal current in the emitters (i_e) and the signal voltage v_{be} for each BJT.
- What is the total emitter current in each BJT?
- What is the signal voltage at each collector? Assume $\alpha = 1$.
- What is the voltage gain realized when the output is taken between the two collectors?

D 9.42 Design a BJT differential amplifier to amplify a differential input signal of 0.1 V and provide a differential output signal of 2 V. To ensure adequate linearity, it is required to limit the signal amplitude across each base-emitter junction to a maximum of 5 mV. Another design requirement is that the differential input resistance be at least $100\text{ k}\Omega$. The BJTs available are specified to have $\beta \geq 100$. Give the circuit configuration and specify the values of all its components.

D 9.43 Design a bipolar differential amplifier such as that in Fig. 9.18 to operate from $\pm 2.5\text{ V}$ power supplies and to provide differential gain of 60 V/V . The power dissipation in the quiescent state should not exceed 1 mW.

- Specify the values of I and R_C . What dc voltage appears at the collectors?
- If $\beta = 100$, what is the input differential resistance?
- For $v_{id} = 10$ mV, what is the signal voltage at each of the collectors?
- For the situation in (c), what is the maximum allowable value of the input common-mode voltage, V_{CM} ? Recall that to maintain an *nnp* BJT in saturation, v_B should not exceed v_C by more than 0.4 V.

D *9.44 In this problem we explore the trade-off between input common-mode range and differential gain in the design of the bipolar BJT. Consider the bipolar differential amplifier in Fig. 9.14 with the input voltages

$$v_{B1} = V_{CM} + (v_{id}/2)$$

$$v_{B2} = V_{CM} - (v_{id}/2)$$

- Bearing in mind that for a BJT to remain in the active mode, v_{BC} should not exceed 0.4 V, show that when v_{id} has a peak $\hat{v}_{id}$, the maximum input common-mode voltage V_{CMmax} is given by

$$V_{CMmax} = V_{CC} + 0.4 - \frac{\hat{v}_{id}}{2} - A_d \left(V_T + \frac{\hat{v}_{id}}{2} \right)$$

- For the case $V_{CC} = 2.5$ V and $\hat{v}_{id} = 10$ mV, use the relationship above to determine V_{CMmax} for the case $A_d = 50$ V/V. Also find the peak output signal $\hat{v}_{od}$ and the required value of IR_C . Now if the power dissipation in the circuit is to be limited to 1 mW in the quiescent state (i.e., with $v_{id} = 0$), find I and R_C . (Remember to include the power drawn from the negative power supply $-V_{EE} = -2.5$ V.)
- If V_{CMmax} is to be +1 V, and all other conditions remain the same, what maximum gain A_d is achievable?

9.45 For the differential amplifier of Fig. 9.14, let $V_{CC} = +5$ V and $IR_C = 4$ V. Find the differential gain A_d . Sketch and clearly label the waveforms for the total collector voltages v_{C1} and v_{C2} and for $(v_{C2} - v_{C1})$ for the case:

$$v_{B1} = 1 + 0.005 \sin(\omega t)$$

$$v_{B2} = 1 - 0.005 \sin(\omega t)$$

9.46 Consider a bipolar differential amplifier in which the collector resistors R_C are replaced with simple current sources implemented using *pnp* transistors. Sketch the circuit and give its differential half-circuit. If $V_A = 20$ V for all transistors, find the differential voltage gain achieved.

9.47 For each of the emitter-degenerated differential amplifiers shown in Fig. P9.47, find the differential half-circuit and derive expressions for the differential gain A_d and differential input resistance R_{id} . For each circuit, what dc voltage appears across the bias current source(s) in the quiescent state (i.e., with $v_{id} = 0$)? Hence, which of the two circuits will allow a larger negative V_{CM} ?

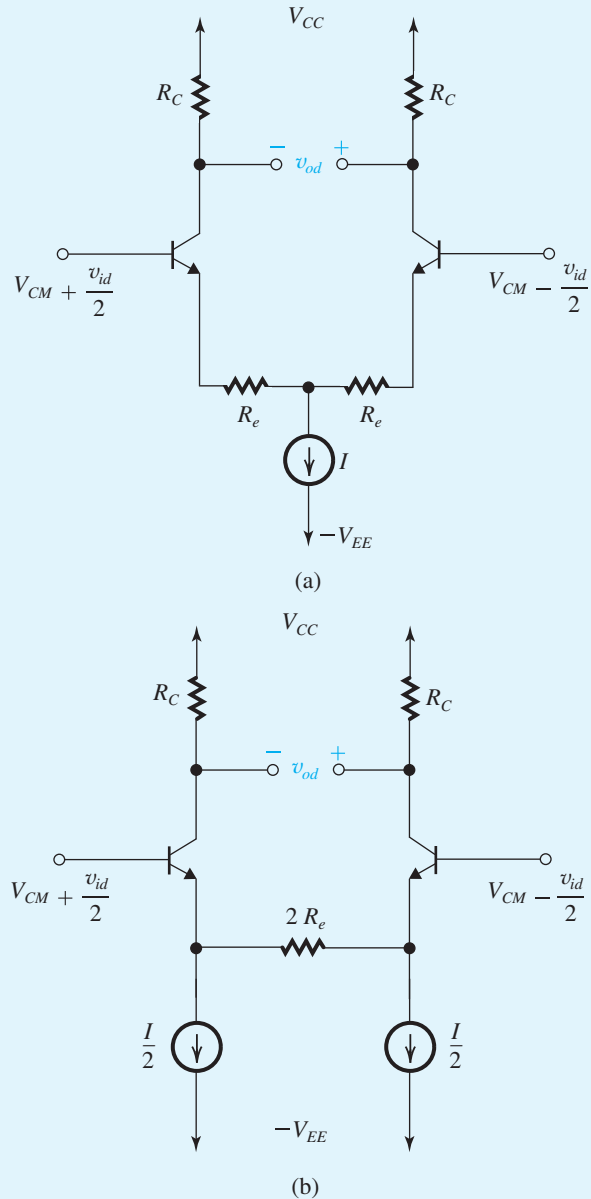


Figure P9.47

9.48 Consider a bipolar differential amplifier that, in addition to the collector resistances R_C , has a load resistance R_L connected between the two collectors. What does the differential gain A_d become?

9.49 A bipolar differential amplifier having resistance R_e inserted in series with each emitter (as in Fig. P9.47(a)) is biased with a constant current I . When both input terminals are grounded, the dc voltage measured across each R_e is found to be $4 V_T$ and that measured across each R_C is found to be $60 V_T$. What differential voltage gain A_d do you expect the amplifier to have?

9.50 A bipolar differential amplifier with emitter-degeneration resistances R_e and R_e is fed with the arrangement shown in Fig. P9.50. Derive an expression for the overall differential voltage gain $G_v \equiv v_{od}/v_{sig}$. If R_{sig} is of such a value that $v_{id} = 0.5v_{sig}$, find the gain G_v in terms of R_C , r_e , R_e , and α . Now if β is doubled, by what factor does G_v increase?

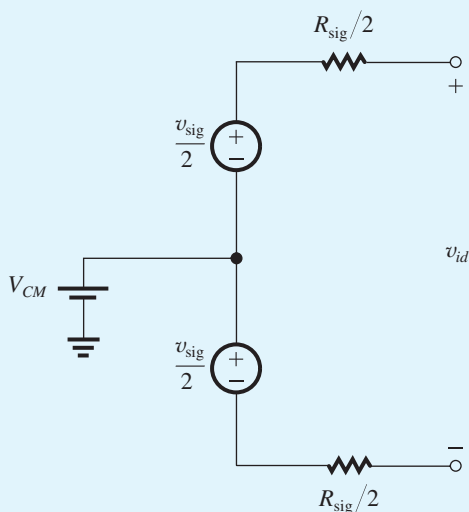


Figure P9.50

9.51 A particular differential amplifier operates from an emitter current source $I = 0.4$ mA. Each of the collector resistances $R_C = 20$ k Ω and a load resistance $R_L = 40$ k Ω is connected between the two collectors. If the amplifier is fed in the manner shown in Fig. P9.50 with $R_{sig} = 100$ k Ω , find the overall voltage gain. Let $\beta = 100$.

9.52 Find the voltage gain and the input resistance of the amplifier shown in Fig. P9.52 assuming $\beta = 100$.

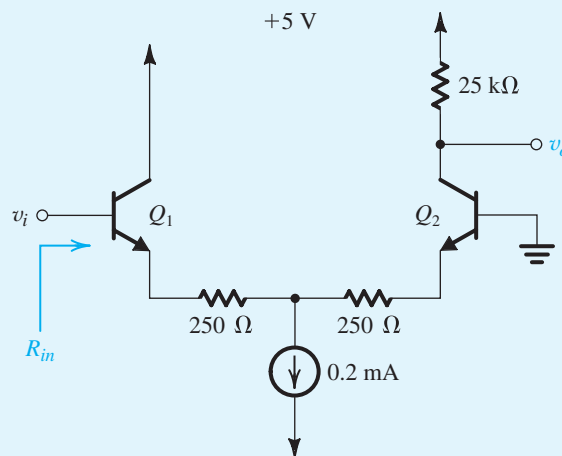


Figure P9.52

9.53 Find the voltage gain and input resistance of the amplifier in Fig. P9.53 assuming that $\beta = 100$.

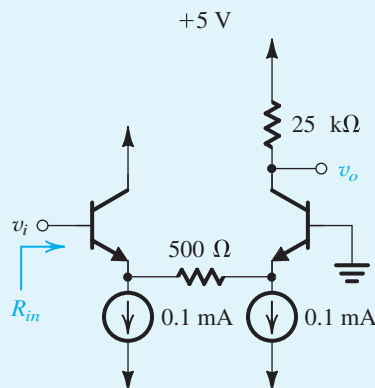


Figure P9.53

9.54 Derive an expression for the small-signal voltage gain v_o/v_i of the circuit shown in Fig. P9.54 in two different ways:

- as a differential amplifier
- as a cascade of a common-collector stage Q_1 and a common-base stage Q_2

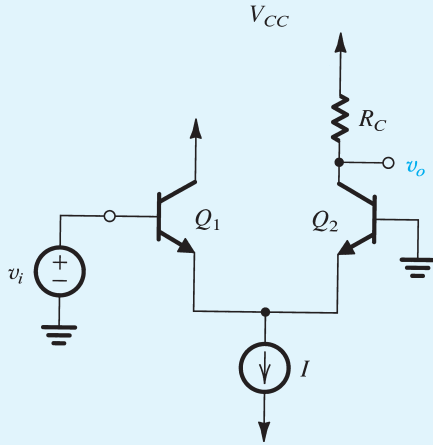


Figure P9.54

Assume that the BJTs are matched and have a current gain α , and neglect the Early effect. Verify that both approaches lead to the same result.

Section 9.3: Common-Mode Rejection

SIM 9.55 An NMOS differential pair is biased by a current source $I = 0.2$ mA having an output resistance $R_{SS} = 100$ k Ω . The amplifier has drain resistances $R_D = 10$ k Ω , using transistors with $k'_n W/L = 3$ mA/V², and r_o that is large. If the output is taken differentially and there is a 1% mismatch between the drain resistances, find $|A_d|$, $|A_{cm}|$, and CMRR.

9.56 For the differential amplifier shown in Fig. P9.2, let Q_1 and Q_2 have $k'_p(W/L) = 4$ mA/V², and assume that the bias current source has an output resistance of 30 k Ω . Find $|V_{ov}|$, g_m , $|A_d|$, $|A_{cm}|$, and the CMRR (in dB) obtained with the output taken differentially. The drain resistances are known to have a mismatch of 2%.

SIM D *9.57 The differential amplifier in Fig. P9.57 utilizes a resistor R_{SS} to establish a 1-mA dc bias current. Note that this amplifier uses a single 5-V supply and thus the dc common-mode voltage V_{CM} cannot be zero. Transistors Q_1 and Q_2 have $k'_n W/L = 2.5$ mA/V², $V_t = 0.7$ V, and $\lambda = 0$.

- Find the required value of V_{CM} .
- Find the value of R_D that results in a differential gain A_d of 8 V/V.
- Determine the dc voltage at the drains.
- Determine the single-ended-output common-mode gain $\Delta V_{D1}/\Delta V_{CM}$. (Hint: You need to take $1/g_m$ into account.)

- Use the common-mode gain found in (d) to determine the change in V_{CM} that results in Q_1 and Q_2 entering the triode region.

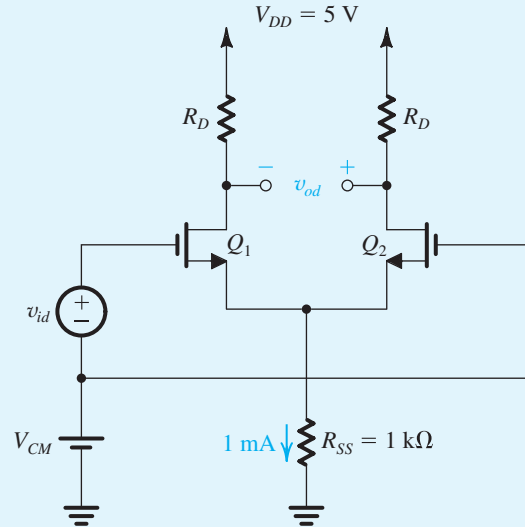


Figure P9.57

9.58 It can be shown that if the drain resistors of a MOS differential amplifier have a mismatch ΔR_D and if simultaneously the transconductances of Q_1 and Q_2 have a mismatch Δg_m , the common-mode gain is given by

$$A_{cm} \simeq \left(\frac{R_D}{2R_{SS}} \right) \left(\frac{\Delta g_m}{g_m} + \frac{\Delta R_D}{R_D} \right)$$

Note that this equation indicates that R_D can be deliberately varied to compensate for the initial variability in g_m and R_D , that is, to minimize A_{cm} .

In a MOS differential amplifier for which $R_D = 5$ k Ω and $R_{SS} = 25$ k Ω , the common-mode gain is measured and found to be 0.002 V/V. Find the percentage change required in one of the two drain resistors so as to reduce A_{cm} to zero (or close to zero).

D 9.59 It is required to design a MOS differential amplifier to have a CMRR of 80 dB. The only source of mismatch in the circuit is a 2% difference between the W/L ratios of the two transistors. Let $I = 100$ μ A and assume that all transistors are operated at $V_{ov} = 0.2$ V. For the 0.18- μ m CMOS fabrication process available, $V'_A = 5$ V/ μ m. What is the value of L required for the current-source transistor?

D 9.60 A MOS differential amplifier utilizing a simple current source to provide the bias current I is found to have a CMRR of 60 dB. If it is required to raise the CMRR to 100 dB by adding a cascode transistor to the current source, what must the intrinsic gain A_0 of the cascode transistor be? If the cascode transistor is operated at $V_{OV} = 0.2$ V, what must its V_A be? If for the specific technology utilized $V'_A = 5$ V/ μm , specify the channel length L of the cascode transistor.

9.61 The differential amplifier circuit of Fig. P9.61 utilizes a resistor connected to the negative power supply to establish the bias current I .

- For $v_{B1} = v_{id}/2$ and $v_{B2} = -v_{id}/2$, where v_{id} is a small signal with zero average, find the magnitude of the differential gain, $|v_o/v_{id}|$.
- For $v_{B1} = v_{B2} = v_{icm}$, where v_{icm} has a zero average, find the magnitude of the common-mode gain, $|v_o/v_{icm}|$.
- Calculate the CMRR.
- If $v_{B1} = 0.1 \sin 2\pi \times 60t + 0.005 \sin 2\pi \times 1000t$, volts, and $v_{B2} = 0.1 \sin 2\pi \times 60t - 0.005 \sin 2\pi \times 1000t$, volts, find v_o .

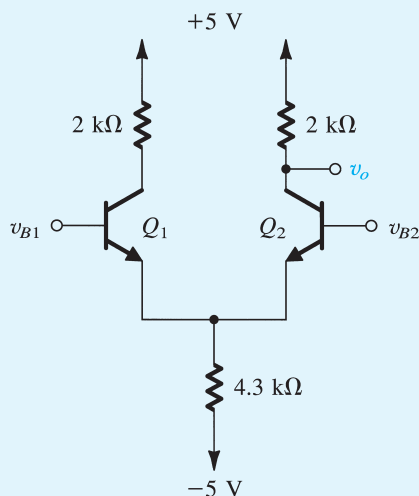


Figure P9.61

9.62 For the differential amplifier shown in Fig. P9.62, identify and sketch the differential half-circuit and the common-mode half-circuit. Find the differential gain, the differential input resistance, the common-mode gain assuming the resistances R_C have 1% tolerance, and the common-mode

input resistance. For these transistors, $\beta = 100$ and $V_A = 100$ V.

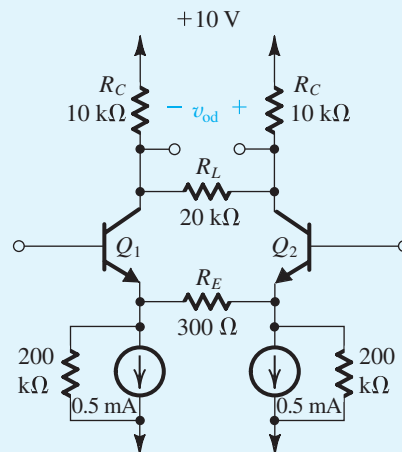


Figure P9.62

9.63 Consider the basic differential circuit in which the transistors have $\beta = 100$ and $V_A = 100$ V, with $I = 0.2$ mA, $R_{EE} = 500$ kΩ, and $R_C = 25$ kΩ. The collector resistances are matched to within 1%. Find:

- the differential gain
- the differential input resistance
- the common-mode gain
- the common-mode rejection ratio
- the common-mode input resistance

9.64 In a bipolar differential-amplifier circuit, the bias current generator consists of a simple common-emitter transistor operating at $200 \mu\text{A}$. For this transistor, and those used in the differential pair, $V_A = 20$ V and $\beta = 50$. What common-mode input resistance would result? Assume $R_C \ll r_o$.

9.65 A bipolar differential amplifier with $I = 0.5$ mA utilizes transistors for which $V_A = 50$ V and $\beta = 100$. The collector resistances $R_C = 5$ kΩ and are matched to within 10%. Find:

- the differential gain
- the common-mode gain and the CMRR if the bias current I is generated using a simple current mirror
- the common-mode gain and the CMRR if the bias current I is generated using a Wilson mirror. (Refer to Eq. 8.95 for R_o of the Wilson mirror.)

D 9.66 It is required to design a differential amplifier to provide the largest possible signal to a pair of 10-kΩ load

resistances. The input differential signal is a sinusoid of 5-mV peak amplitude, which is applied to one input terminal while the other input terminal is grounded. The power supply V_{CC} available is 5 V. To determine the required bias current I , derive an expression for the total voltage at each of the collectors in terms of V_{CC} and I in the presence of the input signal. Then impose the condition that both transistors remain well out of saturation with a minimum v_{CB} of approximately 0 V. Thus determine the required value of I . For this design, what differential gain is achieved? What is the amplitude of the signal voltage obtained between the two collectors? Assume $\alpha \simeq 1$.

D *9.67 Design a BJT differential amplifier that provides two single-ended outputs (at the collectors). The amplifier is to have a differential gain (to each of the two outputs) of at least 100 V/V, a differential input resistance ≥ 10 k Ω , and a common-mode gain (to each of the two outputs) no greater than 0.1 V/V. Use a 2-mA current source for biasing. Give the complete circuit with component values and suitable power supplies that allow for ± 2 V swing at each collector. Specify the minimum value that the output resistance of the bias current source must have. If the current source is realized by a simple mirror, what must the minimum value of V_A be? The BJTs available have $\beta \geq 100$. What is the value of the input common-mode resistance when the bias source has the lowest acceptable output resistance?

9.68 When the output of a BJT differential amplifier is taken differentially, its CMRR is found to be 34 dB higher than when the output is taken single-endedly. If the only source of common-mode gain when the output is taken differentially is the mismatch in collector resistances, what must this mismatch be (in percent)?

***9.69** In a particular BJT differential amplifier, a production error results in one of the transistors having an emitter–base junction area that is twice that of the other. With the inputs grounded, how will the emitter bias current split between the two transistors? If the output resistance of the current source is 500 k Ω and the resistance in each collector (R_C) is 12 k Ω , find the common-mode gain obtained when the output is taken differentially. Assume $\alpha \simeq 1$. [Hint: The CM signal current v_{icm}/R_{EE} will split between Q_1 and Q_2 in the same ratio as the bias current I does.]

Section 9.4: DC Offset

D 9.70 An NMOS differential pair is to be used in an amplifier whose drain resistors are 10 k $\Omega \pm 1\%$. For the

pair, $k'_n W/L = 4$ mA/V². A decision is to be made concerning the bias current I to be used, whether 160 μ A or 360 μ A. Contrast the differential gain and input offset voltage for the two possibilities.

D 9.71 An NMOS differential amplifier for which the MOSFETs have a transconductance parameter k_n and whose drain resistances R_D have a mismatch ΔR_D is biased with a current I .

- Find expressions for A_d and V_{OS} in terms of k_n , R_D , $\Delta R_D/R_D$, and I . Use these expressions to relate V_{OS} and A_d .
- If $k_n = 4$ mA/V², $R_D = 10$ k Ω , and $\Delta R_D/R_D = 0.02$, find the maximum gain realized if V_{OS} is to be limited to 1 mV, 2 mV, 3 mV, 4 mV, and 5 mV. For each case, give the value of the required bias current I . Note the trade-off between gain and offset voltage.

D 9.72 An NMOS amplifier, whose designed operating point is at $V_{OV} = 0.3$ V, is suspected to have a variability of V_t of ± 5 mV, and of W/L and R_D (independently) of $\pm 1\%$. What is the worst-case input offset voltage you would expect to find? What is the major contribution to this total offset? If you used a variation of one of the drain resistors to reduce the output offset to zero and thereby compensate for the uncertainties (including that of the other R_D), what percentage change from nominal would you require?

9.73 An NMOS differential pair operating at a bias current I of 100 μ A uses transistors for which $k'_n = 200$ μ A/V² and $W/L = 10$. Find the three components of input offset voltage under the conditions that $\Delta R_D/R_D = 4\%$, $\Delta(W/L)/(W/L) = 4\%$, and $\Delta V_t = 5$ mV. In the worst case, what might the total offset be? For the usual case of the three effects being independent, what is the offset likely to be?

9.74 A bipolar differential amplifier uses two well-matched transistors, but collector load resistors that are mismatched by 10%. What input offset voltage is required to reduce the differential output voltage to zero?

9.75 A bipolar differential amplifier uses two transistors whose scale currents I_S differ by 10%. If the two collector resistors are well matched, find the resulting input offset voltage.

9.76 Modify Eq. (9.114) for the case of a differential amplifier having a resistance R_E connected in the emitter of each transistor. Let the bias current source be I .

9.77 A differential amplifier uses two transistors whose β values are β_1 and β_2 . If everything else is matched, show that

the input offset voltage is approximately $V_T[(1/\beta_1) - (1/\beta_2)]$. Evaluate V_{OS} for $\beta_1 = 50$ and $\beta_2 = 100$.

9.78 Two possible differential amplifier designs are considered, one using BJTs and the other MOSFETs. In both cases, the collector (drain) resistors are maintained within $\pm 2\%$ of nominal value. The MOSFETs are operated at $V_{OV} = 200$ mV. What input offset voltage results in each case? What does the MOS V_{OS} become if the devices are increased in width by a factor of 4 while the bias current is kept constant?

***9.79** A differential amplifier uses two transistors having V_A values of 100 V and 200 V. If everything else is matched, find the resulting input offset voltage. Assume that the two transistors are intended to be biased at a V_{CE} of about 10 V.

***9.80** A differential amplifier is fed in a balanced or push-pull manner, and the source resistance in series with each base is R_s . Show that a mismatch ΔR_s between the values of the two source resistances gives rise to an input offset voltage of approximately $(I/2\beta)\Delta R_s / [1 + (g_m R_s)/\beta]$.

9.81 One approach to “offset correction” involves the adjustment of the values of R_{C1} and R_{C2} so as to reduce the differential output voltage to zero when both input terminals are grounded. This offset-nulling process can be accomplished by utilizing a potentiometer in the collector circuit, as shown in Fig. P9.81. We wish to find the

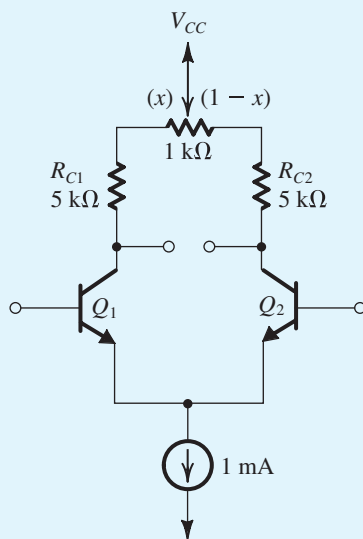


Figure P9.81

potentiometer setting, represented by the fraction x of its value connected in series with R_{C1} , that is required for nulling the output offset voltage that results from:

- (a) R_{C1} being 4% higher than nominal and R_{C2} 4% lower than nominal
- (b) Q_1 having an area 5% larger than nominal, while Q_2 has area 5% smaller than nominal.

9.82 A differential amplifier for which the total emitter bias current is 400 μ A uses transistors for which β is specified to lie between 80 and 200. What is the largest possible input bias current? The smallest possible input bias current? The largest possible input offset current?

****9.83** In a particular BJT differential amplifier, a production error results in one of the transistors having an emitter-base junction area twice that of the other. With both inputs grounded, find the current in each of the two transistors and hence the dc offset voltage at the output, assuming that the collector resistances are equal. Use small-signal analysis to find the input voltage that would restore current balance to the differential pair. Repeat using large-signal analysis and compare results.

D 9.84 A large fraction of mass-produced differential-amplifier modules employing 20-k Ω collector resistors is found to have an input offset voltage ranging from +2 mV to -2 mV. By what amount must one collector resistor be adjusted to reduce the input offset to zero? If an adjustment mechanism is devised that raises one collector resistance while correspondingly lowering the other, what resistance change is needed? If a potentiometer connected as shown in Fig. P9.81 is used, what value of potentiometer resistance (specified to 1 significant digit) is needed? Assume that the offset is entirely due to the finite tolerance of R_C .

Section 9.5: The Differential Amplifier with a Current-Mirror Load

9.85 The differential amplifier of Fig. 9.32(a) is measured and found to have a short-circuit transconductance of 2 mA/V. A differential input signal is applied and the output voltage is measured with a load resistance R_L connected. It is found that when R_L is reduced from ∞ to 20 k Ω , the magnitude of the output signal is reduced by half. What do you estimate R_o and A_d (with R_L disconnected) to be?

9.86 A current-mirror-loaded NMOS differential amplifier is fabricated in a technology for which $|V'_A| = 5$ V/ μ m. All the

transistors have $L = 0.5 \mu\text{m}$. If the differential-pair transistors are operated at $V_{OV} = 0.25 \text{ V}$, what open-circuit differential gain is realized?

9.87 The differential amplifier of Fig. 9.32(a) is biased with $I = 200 \mu\text{A}$. All transistors have $L = 0.5 \mu\text{m}$, and Q_1 and Q_2 have $W/L = 50$. The circuit is fabricated in a process for which $\mu_n C_{ox} = 200 \mu\text{A/V}^2$ and $|V_A| = 5 \text{ V}/\mu\text{m}$. Find $g_{m1,2}$, r_{o2} , r_{o4} , and A_d .

D 9.88 In a current-mirror-loaded differential amplifier of the form shown in Fig. 9.32(a), all transistors are characterized by $k'W/L = 4 \text{ mA/V}^2$, and $|V_A| = 5 \text{ V}$. Find the bias current I for which the gain $v_o/v_{id} = 20 \text{ V/V}$.

D 9.89 Consider a current-mirror-loaded differential amplifier such as that shown in Fig. 9.32(a) with the bias current source implemented with the modified Wilson mirror of Fig. P9.89 with $I = 200 \mu\text{A}$. The transistors have $|V_t| = 0.5 \text{ V}$ and $k'W/L = 5 \text{ mA/V}^2$. What is the lowest value of the total power supply ($V_{DD} + V_{SS}$) that allows each transistor to operate with $|V_{DS}| \geq |V_{GS}|$?

***9.90** (a) Sketch the circuit of a current-mirror-loaded MOS differential amplifier in which the input transistors are cascoded and a cascode current mirror is used for the load.

(b) Show that if all transistors are operated at an overdrive voltage V_{OV} and have equal Early voltages $|V_A|$, the gain is given by

$$A_d = 2(V_A/V_{OV})^2$$

Evaluate the gain for $V_{OV} = 0.20 \text{ V}$ and $V_A = 10 \text{ V}$.

9.91 Figure P9.91 shows the current-mirror-loaded MOS differential amplifier prepared for small-signal analysis. We have “pulled out” r_o of each transistor; thus, the current in the drain of each transistor will be $g_m v_{gs}$. To help the reader, we have already indicated approximate values for some of the node voltages. For instance, the output voltage $v_o = \frac{1}{2}(g_m r_o) v_{id}$, which we have derived in the text. The voltage at the common sources has been found to be approximately $+v_{id}/4$, which is very far from the virtual ground one might assume. Also, the voltage at the gate of the mirror is approximately $-v_{id}/4$, confirming our contention that the voltage there is vastly different from the output voltage, hence the lack of balance in the circuit and the unavailability of a differential half-circuit. Find the currents labeled i_1 to i_{13} in terms of $(g_m v_{id})$. Determine their values in the sequence of their numbering and assume $g_m r_o \gg 1$. Note that all transistors are assumed to be operating at the same $|V_{OV}|$. Write the current values on the circuit diagram and reflect on the results.

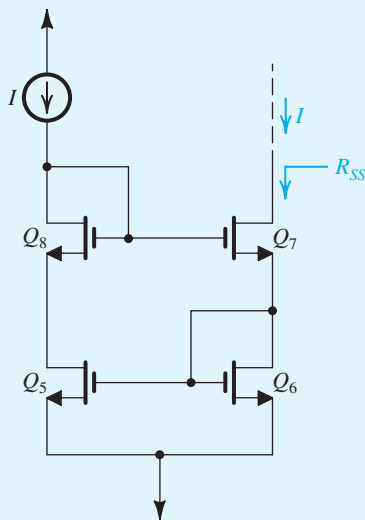


Figure P9.89

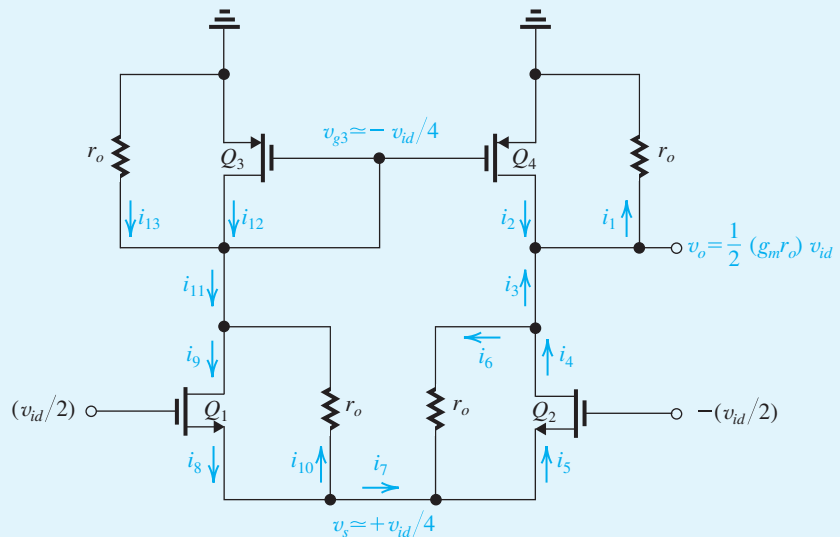


Figure P9.91

9.92 A current-mirror-loaded NMOS differential amplifier operates with a bias current I of 200 μA . The NMOS transistors are operated at $V_{OV} = 0.2\text{ V}$ and the PMOS devices at $|V_{OV}| = 0.3\text{ V}$. The Early voltages are 20 V for the NMOS and 12 V for the PMOS transistors. Find G_m , R_o , and A_d . For what value of load resistance is the gain reduced by a factor of 2?

9.93 This problem investigates the effect of transistor mismatches on the input offset voltage of the current-mirror-loaded MOS differential amplifier of Fig. 9.32(a). For this purpose, ground both input terminals and short-circuit the output node to ground.

(a) If the amplifying transistors Q_1 and Q_2 exhibit a W/L mismatch of $\Delta(W/L)_A$, find the resulting short-circuit output current and hence show that the corresponding V_{OS} is given by

$$V_{OS1} = (V_{OV}/2) \frac{\Delta(W/L)_A}{(W/L)_A}$$

where V_{OV} is the overdrive voltage at which Q_1 and Q_2 are operating.

(b) Repeat for a mismatch $\Delta(W/L)_M$ in the W/L ratios of the mirror transistor Q_3 and Q_4 to show that the corresponding V_{OS} is given by

$$V_{OS2} = (V_{OV}/2) \frac{\Delta(W/L)_M}{(W/L)_M}$$

where V_{OV} is the overdrive voltage at which Q_1 and Q_2 are operating.

(c) For a circuit in which all transistors are operated at $|V_{OV}| = 0.2\text{ V}$ and all W/L ratios are accurate to within $\pm 1\%$ of nominal, find the worst-case total offset voltage V_{OS} .

9.94 The differential amplifier in Fig. 9.36(a) is operated with $I = 500\text{ }\mu\text{A}$, with devices for which $V_A = 10\text{ V}$ and $\beta = 100$. What differential input resistance, output resistance, short-circuit transconductance, and open-circuit voltage gain would you expect? What will the voltage gain be if the input resistance of the subsequent stage is equal to R_{id} of this stage?

9.95 A bipolar differential amplifier having a simple pnp current-mirror load is found to have an input offset voltage of 2 mV. If the offset is attributable entirely to the finite β of the pnp transistors, what must β_p be?

9.96 For the current-mirror-loaded bipolar differential pair, replacing the simple current-mirror load by the base-current-compensated mirror of Fig. 8.11, find the expected systematic input offset voltage. Evaluate V_{OS} for $\beta_p = 50$.

9.97 For the current-mirror-loaded bipolar differential pair, replacing the simple current-mirror load by the Wilson mirror of Fig. 8.40(a), find the expected systematic input offset voltage. Evaluate V_{OS} for $\beta_p = 50$.

9.98 Figure P9.98 shows a differential cascode amplifier with an active load formed by a Wilson current mirror. Utilizing the expressions derived in Chapter 8 for the output resistance of a bipolar cascode and the output resistance of the Wilson mirror, and assuming all transistors to be identical, show that the differential voltage gain A_d is given approximately by

$$A_d = \frac{1}{3} \beta g_m r_o$$

Evaluate A_d for the case of $\beta = 100$ and $V_A = 20\text{ V}$.

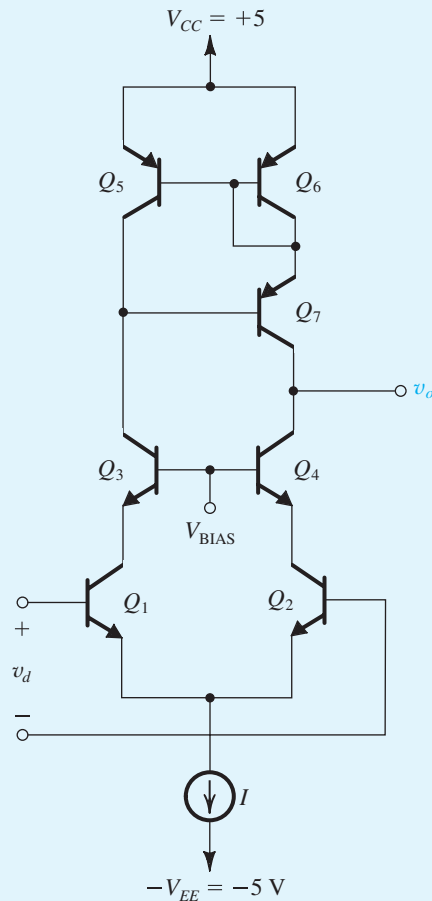


Figure P9.98

D 9.99 Consider the bias design of the Wilson-loaded cascode differential amplifier shown in Fig. P9.98.

- What is the largest signal voltage possible at the output without Q_7 saturating? Assume that the CB junction conducts when the voltage across it exceeds 0.4 V.
- What should the dc bias voltage established at the output (by an arrangement not shown) be in order to allow for positive output signal swing of 1.5 V?
- What should the value of V_{BIAS} be in order to allow for a negative output signal swing of 1.5 V?
- What is the upper limit on the input common-mode voltage v_{CM} ?

****9.100** Figure P9.100 shows a modified cascode differential amplifier. Here Q_3 and Q_4 are the cascode transistors. However, the manner in which Q_3 is connected with its base current feeding the current mirror Q_7 – Q_8 results in very interesting input properties. Note that for simplicity the circuit is shown with the base of Q_2 grounded. Assume that all transistors have equal β 's.

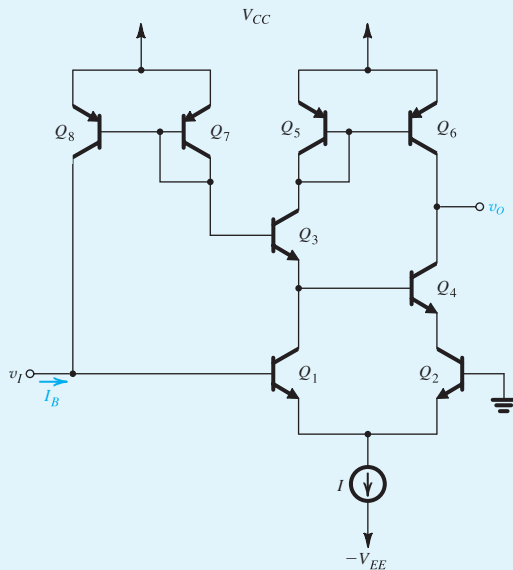


Figure P9.100

- With $v_i = 0$ V dc, find the input bias current I_B assuming all transistors have equal value of β . Compare with the case without the Q_7 – Q_8 connection.

- With $v_i = 0$ V (dc) + v_{id} , find the input signal current i_i and hence the input differential resistance R_{id} . Compare with the case without the Q_7 – Q_8 connection. By what factor does R_{id} increase?

9.101 For the folded-cascode differential amplifier of Fig. 9.38, find the value of V_{BIAS} that results in the largest possible positive output swing, while keeping Q_3 , Q_4 , and the *pnp* transistors that realize the current sources out of saturation. Assume $V_{\text{CC}} = V_{\text{EE}} = 5$ V. If the dc level at the output is 0 V, find the maximum allowable output signal swing. For $I = 0.5$ mA, $\beta_P = 50$, $\beta_N = 100$, and $V_A = 100$ V find G_m , R_{o4} , R_{o5} , R_o , and A_d .

9.102 For the BiCMOS differential amplifier in Fig. P9.102 let $V_{\text{DD}} = V_{\text{SS}} = 3$ V, $I = 0.2$ mA, $k'_p W/L = 6.4$ mA/V²; $|V_A|$ for *p*-channel MOSFETs is 10 V, $|V_A|$ for *npn* transistors is 30 V. Find G_m , R_o , and A_d .

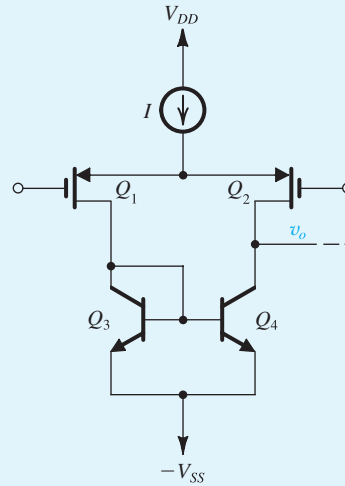


Figure P9.102

SIM D 9.103 It is required to design the current-mirror-loaded differential MOS amplifier of Fig. 9.32 to obtain a differential gain of 50 V/V. The technology available provides $\mu_n C_{ox} = 4\mu_p C_{ox} = 400$ $\mu\text{A/V}^2$, $|V_t| = 0.5$ V, and $|V'_A| = 20$ V/ μm and operates from ± 1 V supplies. Use a bias current $I = 200$ μA and operate all devices at $|V_{\text{OV}}| = 0.2$ V.

- Find the W/L ratios of the four transistors.
- Specify the channel length required of all transistors.
- If $V_{\text{ICM}} = 0$, what is the allowable range of v_o ?

- (d) If I is delivered by a simple NMOS current source operated at the same V_{OV} and having the same channel length as the other four transistors, determine the CMRR obtained.

9.104 Consider the current-mirror-loaded MOS differential amplifier of Fig. 9.32(a) in two cases:

- (a) Current source I is implemented with a simple current mirror.
 (b) Current source I is implemented with the modified Wilson current mirror shown in Fig. P9.104.

Recalling that for the simple mirror $R_{SS} = r_{o1} \parallel r_{o5}$ and for the Wilson mirror $R_{SS} \simeq g_{m7} r_{o7} r_{o5}$, and assuming that all transistors have the same $|V_A|$ and $k'W/L$, show that for case (a)

$$\text{CMRR} = 2 \left(\frac{V_A}{V_{OV}} \right)^2$$

and for case (b)

$$\text{CMRR} = 2\sqrt{2} \left(\frac{V_A}{V_{OV}} \right)^3$$

where V_{OV} is the overdrive voltage that corresponds to a drain current of $I/2$. For $k'W/L = 4 \text{ mA/V}^2$, $I = 160 \mu\text{A}$, and $|V_A| = 5 \text{ V}$, find CMRR for both cases.

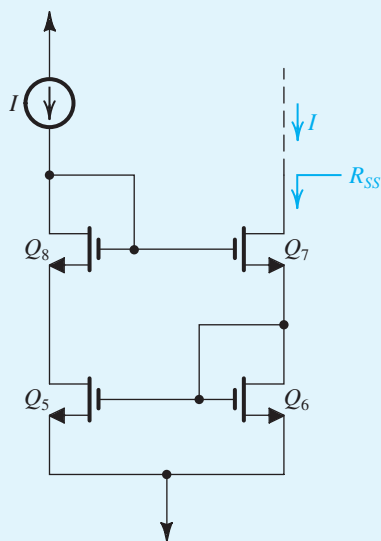


Figure P9.104

9.105 The MOS differential amplifier of Fig. 9.32(a) is biased with a simple current mirror delivering $I = 200 \mu\text{A}$. All transistors are operated at $V_{OV} = 0.2 \text{ V}$ and have $V_A = 5 \text{ V}$. Find G_m , R_o , A_d , R_{SS} , G_{mcm} , R_{im} , A_m , R_{om} , R_{o2} , A_{cm} , and CMRR.

9.106 A current-mirror-loaded MOS differential amplifier is found to have a differential voltage gain A_d of 30 V/V . Its bias current source has an output resistance $R_{SS} = 45 \text{ k}\Omega$. The current mirror utilized has a current gain A_m of 0.98 A/A and an output resistance R_{om} of $45 \text{ k}\Omega$. If the common-mode output resistances of the amplifier, R_{o1} and R_{o2} , are very large, find A_{cm} and CMRR.

9.107 A current-mirror-loaded MOS differential amplifier is found to have a differential voltage gain A_d of 50 V/V and a CMRR of 60 dB . If the output resistance of the bias current source is $20 \text{ k}\Omega$ and the output resistance of the current-mirror load is $20 \text{ k}\Omega$, what is the expected magnitude of the deviation from unity of the current gain of the load mirror?

D *9.108 Design the circuit of Fig. 9.36(a) using a basic current mirror to implement the current source I . It is required that the short-circuit transconductance be 5 mA/V . Use $\pm 5\text{-V}$ power supplies and BJTs that have $\beta = 100$ and $V_A = 100 \text{ V}$. Give the complete circuit with component values and specify the differential input resistance R_{id} , the output resistance R_o , the open-circuit voltage gain A_d , the input bias current, the input common-mode range, the common-mode gain, and the CMRR.

D *9.109 Repeat the design of the amplifier specified in Problem 9.108 utilizing a Widlar current source (Fig. 8.42) to supply the bias current. Assume that the largest resistance available is $2 \text{ k}\Omega$.

9.110 A bipolar differential amplifier such as that shown in Fig. 9.36(a) has $I = 0.4 \text{ mA}$, $V_A = 40 \text{ V}$, and $\beta = 150$. Find G_m , R_o , A_d , and R_{id} . If the bias current source is implemented with a simple $n\text{pn}$ current mirror, find R_{EE} , A_{cm} , and CMRR. If the amplifier is fed differentially with a source having a total of $30 \text{ k}\Omega$ resistance (i.e., $15 \text{ k}\Omega$ in series with the base lead of each of Q_1 and Q_2), find the overall differential voltage gain.

9.111 For the current-mirror-loaded differential pair in Fig. P9.111, find:

- (a) differential input resistance, R_{id}
- (b) A_d
- (c) CMRR

Assume $\beta = 100$, $|V_{BE}| = 0.7$ V, and $|V_A| = 60$ V.

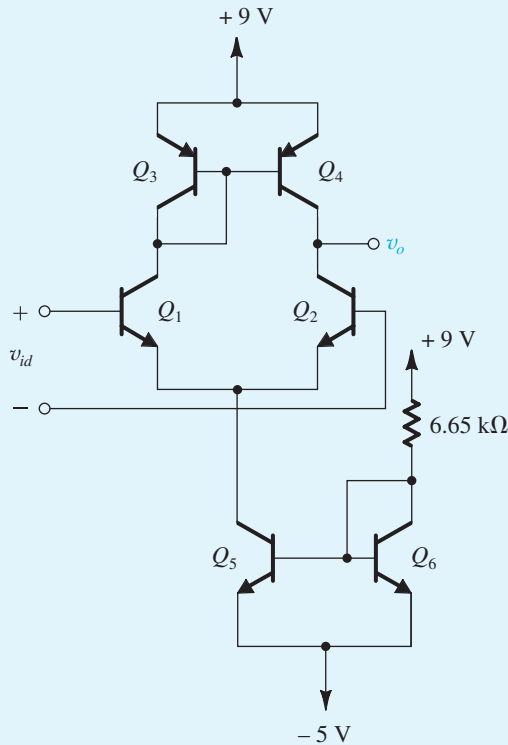


Figure P9.111

9.112 For the current-mirror-loaded differential amplifier in Fig. P9.112, find:

- (a) differential input resistance, R_{id}
- (b) A_d
- (c) CMRR

Assume $\beta = 100$, $|V_{BE}| = 0.7$ V, $|V_A| = 60$ V, $V_t = 0.7$ V, and $k'(W/L) = 2$ mA/V².

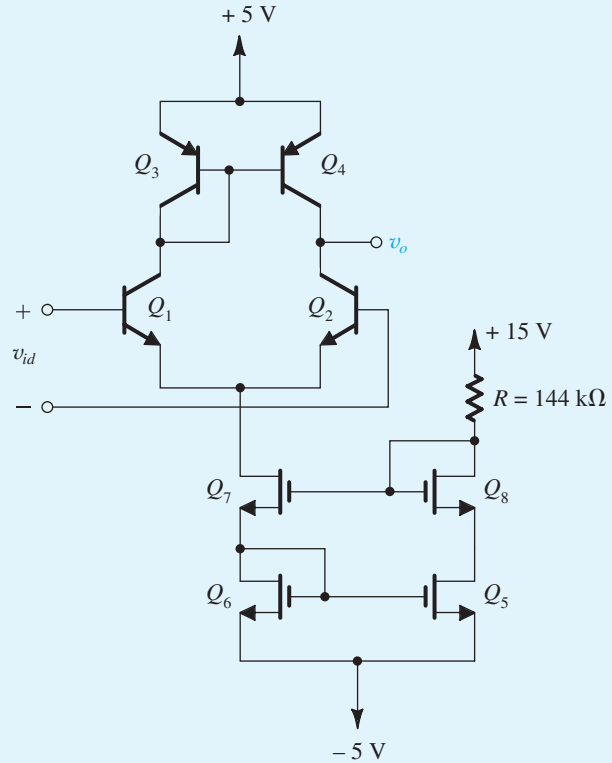


Figure P9.112

Section 9.6: Multistage Amplifiers

9.113 Consider the circuit in Fig. 9.40 with the device geometries (in μm) shown in Table P9.113. Let $I_{\text{REF}} = 225 \mu\text{A}$, $|V_t| = 0.75$ V for all devices, $\mu_n C_{ox} = 180 \mu\text{A/V}^2$, $\mu_p C_{ox} = 60 \mu\text{A/V}^2$, $|V_A| = 9$ V for all devices, $V_{DD} = V_{SS} = 1.5$ V. Determine the width of Q_6 , W , that will ensure that the op amp will not have a systematic offset voltage. Then, for all devices evaluate I_D , $|V_{ov}|$, $|V_{gs}|$, g_m , and r_o . Provide your results in a table similar to Table 9.1. Also find A_1 , A_2 , the open-loop voltage gain, the input

Table P9.113

Transistor	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
W/L	30/0.5	30/0.5	10/0.5	10/0.5	60/0.5	W/0.5	60/0.5	60/0.5

common-mode range, and the output voltage range. Neglect the effect of V_A on the bias currents.

D 9.114 The two-stage CMOS op amp in Fig. P9.114 is fabricated in a $0.18\text{-}\mu\text{m}$ technology having $k'_n = 4k'_p = 400\text{ }\mu\text{A/V}^2$, $V_m = -V_{tp} = 0.4\text{ V}$.

- With A and B grounded, perform a dc design that will result in each of Q_1 , Q_2 , Q_3 , and Q_4 conducting a drain current of $100\text{ }\mu\text{A}$ and each of Q_6 and Q_7 a current of $200\text{ }\mu\text{A}$. Design so that all transistors operate at 0.2-V overdrive voltages. Specify the W/L ratio required for each MOSFET. Present your results in tabular form. What is the dc voltage at the output (ideally)?
- Find the input common-mode range.
- Find the allowable range of the output voltage.

- With $v_A = v_{id}/2$ and $v_B = -v_{id}/2$, find the voltage gain v_o/v_{id} . Assume an Early voltage of 6 V .

D *9.115 In a particular design of the CMOS op amp of Fig. 9.40 the designer wishes to investigate the effects of increasing the W/L ratio of both Q_1 and Q_2 by a factor of 4. Assuming that all other parameters are kept unchanged, refer to Example 9.6 to help you answer the following questions:

- Find the resulting change in $|V_{OV}|$ and in g_m of Q_1 and Q_2 .
- What change results in the voltage gain of the input stage? In the overall voltage gain?
- What is the effect on the input offset voltages? (You might wish to refer to Section 9.4).

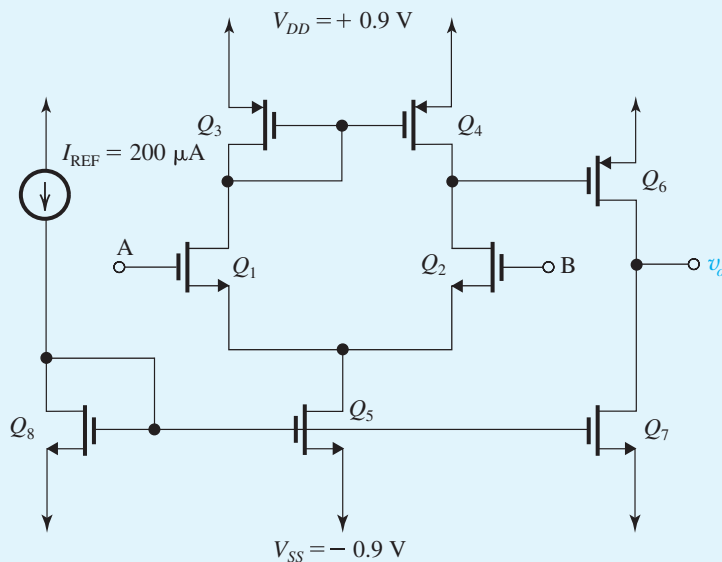


Figure P9.114

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

9.116 Consider the amplifier of Fig. 9.40, whose parameters are specified in Example 9.6. If a manufacturing error results in the W/L ratio of Q_7 being 48/0.8, find the current that Q_7 will now conduct. Thus find the systematic offset voltage that will appear at the output. (Use the results of Example 9.6.) Assuming that the open-loop gain will remain approximately unchanged from the value found in Example 9.6, find the corresponding value of input offset voltage, V_{os} .

9.117 Consider the input stage of the CMOS op amp in Fig. 9.40 with both inputs grounded. Assume that the two sides of the input stage are perfectly matched except that the threshold voltages of Q_3 and Q_4 have a mismatch ΔV_t . Show that a current $g_{m3}\Delta V_t$ appears at the output of the first stage. What is the corresponding input offset voltage?

9.118 The two-stage op amp in Figure P9.114 is fabricated in a 65-nm technology having $k'_n = 5.4 \times k'_p = 540 \mu\text{A}/\text{V}^2$ and $V_{tn} = -V_{tp} = 0.35 \text{ V}$. The amplifier is operated with $V_{DD} = +1.2 \text{ V}$ and $V_{SS} = 0 \text{ V}$.

- With A and B at a dc voltage of $V_{DD}/2$, perform a dc design that will result in each of Q_1 , Q_2 , Q_3 , and Q_4 conducting a drain current of $200 \mu\text{A}$ and each of Q_6 and Q_7 conducting a current of $400 \mu\text{A}$. Design so that all transistors operate at 0.15-V overdrive voltages. Specify the W/L ratio required for each MOSFET. Present all results in a table.
- Find the input common-mode range.
- Find the allowable range of the output voltage.
- With $v_A = v_{id}/2$ and $v_B = -v_{id}/2$, find the voltage gain v_o/v_{id} . Assume an Early voltage of 1.8 V.

***9.119** Figure P9.119 shows a bipolar op-amp circuit that resembles the CMOS op amp of Fig. 9.40. Here, the input differential pair Q_1 – Q_2 is loaded in a current mirror formed by Q_3 and Q_4 . The second stage is formed by the current-source-loaded common-emitter transistor Q_5 . Unlike the CMOS circuit, here there is an output stage formed by the emitter follower Q_6 . The function of capacitor C_C will be explained later, in Chapter 11. All transistors have $\beta = 100$, $|V_{BE}| = 0.7 \text{ V}$, and $r_o = \infty$.

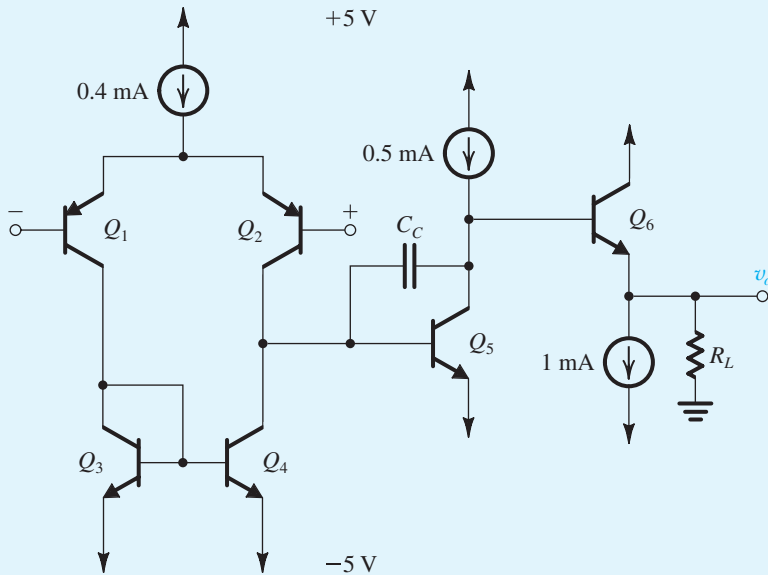


Figure P9.119

- (a) For inputs grounded and output held at 0 V (by negative feedback, not shown) find the emitter currents of all transistors.
- (b) Calculate the gain of the amplifier with $R_L = 1 \text{ k}\Omega$.

9.120 A BJT differential amplifier, biased to have $r_e = 50 \Omega$ and utilizing two $50\text{-}\Omega$ emitter resistors and $5\text{-k}\Omega$ loads, drives a second differential stage biased to have $r_e = 25 \Omega$. All BJTs have $\beta = 100$. What is the voltage gain of the first stage? Also find the input resistance of the first stage, and the current gain from the input of the first stage to the collectors of the second stage.

9.121 In the multistage amplifier of Fig. 9.41, emitter resistors are to be introduced— 100Ω in the emitter lead of each of the first-stage transistors and 25Ω for each of the second-stage transistors. What is the effect on input resistance, the voltage gain of the first stage, and the overall voltage gain? Use the bias values found in Example 9.7.

D 9.122 Consider the circuit of Fig. 9.41 and its output resistance. Which resistor has the most effect on the output resistance? What should this resistor be changed to if the output resistance is to be reduced by a factor of 2? What will the amplifier gain become after this change? What other change can you make to restore the amplifier gain to approximately its prior value?

D 9.123 (a) If, in the multistage amplifier of Fig. 9.41, the resistor R_5 is replaced by a constant-current source $\simeq 1 \text{ mA}$, such that the bias situation is essentially unaffected, what does the overall voltage gain of the amplifier become? Assume that the output resistance of the current source is very high. Use the results of Example 9.8.

(b) With the modification suggested in (a), what is the effect of the change on output resistance? What is the overall gain of the amplifier when loaded by 100Ω to ground? The original amplifier (before modification) has an output resistance of 152Ω and a voltage gain of 8513 V/V . What is its gain when loaded by 100Ω ? Comment. Use $\beta = 100$.

***9.124** Figure P9.124 shows a three-stage amplifier in which the stages are directly coupled. The amplifier, however, utilizes bypass capacitors, and, as such, its frequency response falls off at low frequencies. For our purposes here, we shall assume that the capacitors are large enough to act as perfect short circuits at all signal frequencies of interest.

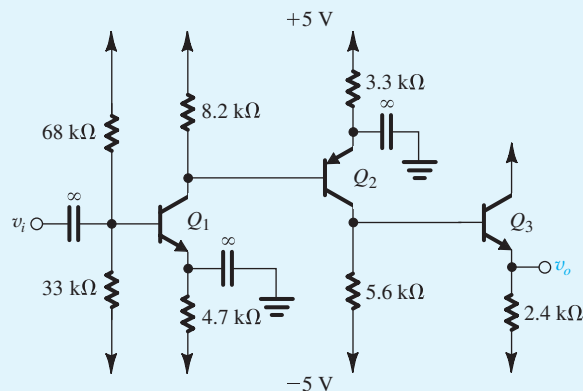


Figure P9.124

- (a) Find the dc bias current in each of the three transistors. Also find the dc voltage at the output. Assume $|V_{BE}| = 0.7 \text{ V}$, $\beta = 100$, and neglect the Early effect.
- (b) Find the input resistance and the output resistance.
- (c) Use the current-gain method to evaluate the voltage gain v_o/v_i .

9.125 For the current mirror in Fig. P9.125, replace the transistors with their hybrid- π models and show that:

$$R_i = \frac{1}{g_{m1}} \parallel r_{o1}$$

$$A_{is} \simeq A_{is}|_{\text{ideal}} \left(1 - \frac{1}{g_{m1} r_{o1}} \right)$$

$$A_{is}|_{\text{ideal}} = g_{m2}/g_{m1}$$

$$R_o = r_{o2}$$

where A_{is} denotes the short-circuit current gain.

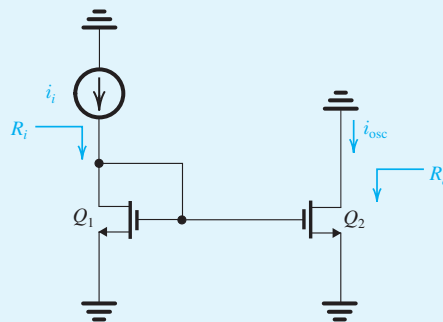


Figure P9.125

****9.126** The MOS differential amplifier shown in Fig. P9.126 utilizes three current mirrors for signal transmission: Q_4 – Q_6 has a transmission factor of 2 [i.e., $(W/L)_6/(W/L)_4 = 2$], Q_3 – Q_5 has a transmission factor of 1, and Q_7 – Q_8 has a transmission factor of 2. All transistors are sized to operate at the same overdrive voltage, $|V_{OV}|$. All transistors have the same Early voltage $|V_A|$.

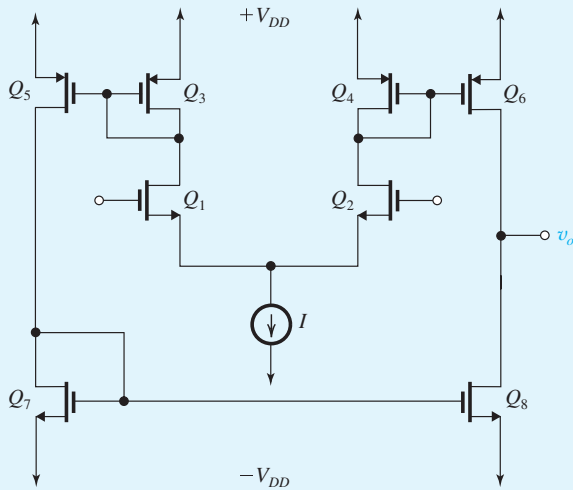


Figure P9.126

- Provide in tabular form the values of I_D , g_m , and r_o of each of the eight transistors in terms of I , V_{OV} , and V_A .
- Show that the differential voltage gain A_d is given by

$$A_d = 2g_{m1}(r_{o6} \parallel r_{o8}) = V_A/V_{OV}$$

- Show that the CM gain is given by

$$|A_{cm}| \simeq \frac{r_{o6} \parallel r_{o8}}{R_{SS}} \frac{1}{g_{m7}r_{o7}}$$

where R_{SS} is the output resistance of the bias current source I . [Hint: Replace each of Q_1 and Q_2 together with their source resistance $2R_{SS}$ with a controlled current-source $v_{icm}/2R_{SS}$ and an output resistance. For each current mirror, the current transfer ratio is given by

$$A_i \simeq A_i(\text{ideal}) \left(1 - \frac{1}{g_m r_o} \right)$$

where g_m and r_o are the parameters of the input transistor of the mirror. (see Problem 9.125 above.)]

- If the current source I is implemented using a simple mirror and the MOS transistor is operated at the same

V_{OV} , show that the CMRR is given by

$$\text{CMRR} = 4(V_A/V_{OV})^2$$

- Find the input CM range and the output linear range in terms of V_{DD} , $|V_t|$, and $|V_{OV}|$.

D *9.127** For the circuit shown in Fig. P9.127, which uses a folded cascode involving transistor Q_3 , all transistors have $|V_{BE}| = 0.7$ V for the currents involved, $V_A = 200$ V, and $\beta = 100$. The circuit is relatively conventional except for Q_5 , which operates in a Class B mode (we will study this in Chapter 12) to provide an increased negative output swing for low-resistance loads.

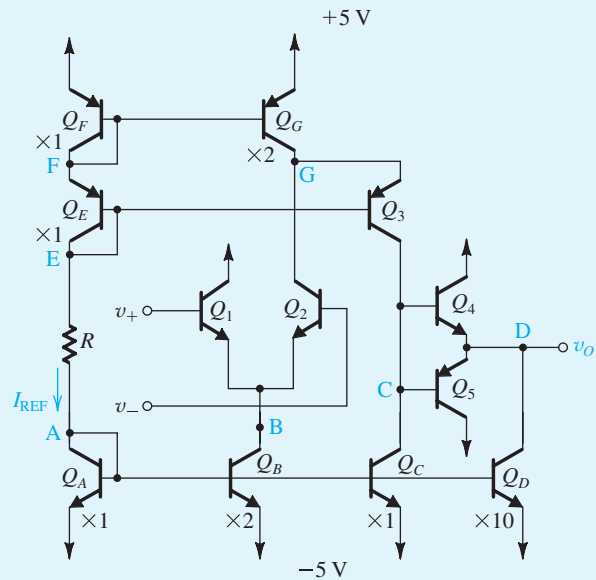


Figure P9.127

- Perform a bias calculation assuming $|V_{BE}| = 0.7$ V, high β , $V_A = \infty$, $v_+ = v_- = 0$ V, and v_o is stabilized by feedback to about 0 V. Find R so that the reference current I_{REF} is $100 \mu\text{A}$. What are the voltages at all the labeled nodes?
- Provide in tabular form the bias currents in all transistors together with g_m and r_o for the signal transistors (Q_1 , Q_2 , Q_3 , Q_4 , and Q_5) and r_o for Q_C , Q_D , and Q_G .
- Now, using $\beta = 100$, find the voltage gain $v_o/(v_+ - v_-)$, and in the process, verify the polarity of the input terminals.

- (d) Find the input and output resistances.
- (e) Find the input common-mode range for linear operation.
- (f) For no load, what is the range of available output voltages, assuming $|V_{CEsat}| = 0.3$ V?
- (g) Now consider the situation with a load resistance connected from the output to ground. At the positive and negative limits of the output signal swing, find the smallest load resistance that can be driven if one or the other of Q_1 or Q_2 is allowed to cut off.

D *9.128** In the CMOS op amp shown in Fig. P9.128, all MOS devices have $|V_t| = 1$ V, $\mu_n C_{ox} = 2 \mu\text{pA/V}^2$, $\mu_p C_{ox} = 40 \mu\text{A/V}^2$, $|V_A| = 50$ V, and $L = 5 \mu\text{m}$. Device widths are indicated on the diagram as multiples of W , where $W = 5 \mu\text{m}$.

- (a) Design R to provide a $10\text{-}\mu\text{A}$ reference current.
- (b) Assuming $v_o = 0$ V, as established by external feedback, perform a bias analysis, finding all the labeled node voltages, and V_{GS} and I_D for all transistors.
- (c) Provide in table form I_D , V_{GS} , g_m , and r_o for all devices.
- (d) Calculate the voltage gain $v_o/(v_+ - v_-)$, the input resistance, and the output resistance.
- (e) What is the input common-mode range?
- (f) What is the output signal range for no load?

- (g) For what load resistance connected to ground is the output negative voltage limited to -1 V before Q_7 begins to conduct?
- (h) For a load resistance one-tenth of that found in (g), what is the output signal swing?

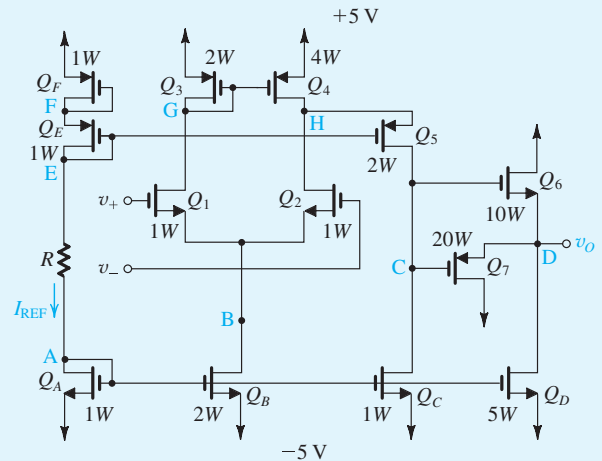


Figure P9.128